2027 International Conference on Artificial Intelligence, Smart Electronic & Energy Systems : Submission (586) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Fri, Oct 16, 2026 at 1:16 PM
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Hello,

The following submission has been edited.

Track Name: Track 5: Signal Processing & VLSI Systems

Paper ID: 586

Paper Title: A CONFIGURABLE PERFORMANCE MONITORING AND ANALYTICS IP FOR RISC-V PROCESSORS

Abstract:
Modern RISC-V processor Performance Moni-toring Units (PMUs) measure a range of events such as branches, cache operations, executed instructions and cycles, and display raw number counts. These metrics need to be manually analyzed by engineers through scripts or a wave-form viewer to provide answers to simple questions such as “Why is the IPC low?”. This PMA IP is a stealthy observer in the RISC-V pipeline, gathering raw information about events in the hardware and making it available as rich performance data without changing the basic design of the hardware. Mi croarchitectural Analysis and Architectural Profiling are two modes of the PMA IP. There are 5 re-usable components of the PMA IP: Event Collector, Event Counters, Analytics En gine, Mode Controller and Register Bank. An example is provided of how the PMA IP can generate derived metrics without any human post processing in the hardware coun ters. The PMA IP can support different RISC-V implementa

tions with analytics. The proposed PMA IP was tested with
benchmark workloads of 10K and 2K instructions and auto
matically collected raw PMU events and derived performance
metrics such as IPC (0.92), CPI (1.07–1.08), 92% pipeline
efficiency and 92% stage utilization, providing these insights
instantaneously and drastically reducing the dependency on
manual post-processing and analysis time. In addition, it ex
plicitly studied the memory operation ratio, hazard distribu
tion and forwarding efficiency (100%) of the hardware. The
results show that the PMA IP can be easily integrated into
a variety of RISC-V processor architectures and is able to
provide accurate architectural and microarchitectural perfor
mance information.
Keywords: RISC-V Processor, Performance Monitoring
Unit (PMU), PMA IP, Microarchitectural Analysis, Archi
tectural Profiling, Hardware Performance Analytics, Pipeline
Processor.

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Submission Files:
RISCV_PROCESSOR_HIGH_PERFORMANCE.pdf (543 Kb, Fri, 16 Oct 2026 07:44:28 GMT)

Submission Questions Response: Not Entered

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